DEPARTMENT OF INFORMATION TECHNOLOGY

SMT. PARMESHWARIDEVI DURGADUTT TIBREWALA

LIONS JUHU COLLEGE

OF ARTS, COMMERCE AND SCIENCE

Affiliated to University of Mumbai

(Autonomous)

J.B. NAGAR, ANDHERI (E), MUMBAI-400059

Academic Year 2025-2026

MODERN NETWORKING

For

Semester II

<u>Submitted By:</u>

Mr. Ahmed Dilshad Shaikh

Msc.IT (Sem II)

SMT. PARMESHWARIDEVI DURGADUTT TIBREWALA

LIONS JUHU COLLEGE

OF ARTS, COMMERCE AND SCIENCE

Affiliated to University of Mumbai

(Autonomous)

J.B. NAGAR, ANDHERI (E), MUMBAI-400059

DEPARTMENT OF INFORMATION TECHNOLOGY

Certificate of Approval

This is to certify that practical entitled **"MODERN NETWORKING"**, Undertaken at **SMT.PARMESHWARIDEVI DURGADUTT TIBREWALA LIONS JUHU COLLEGE OF ARTS, COMMERCE & SCIENCE.**
By **Mr. Ahmed Dilshad Shaikh** **Seat No.** **MIT251009** in partial fulfilment of **M.Sc. (IT) master degree (Semester II)** Examination had not been submitted for any other examination and does not form of any other course undergone by the candidate. It is further certified that he has completed all required phases of the practical.

Internal Examiner **External Examiner**

HOD / In-Charge / Coordinator **Signature/**

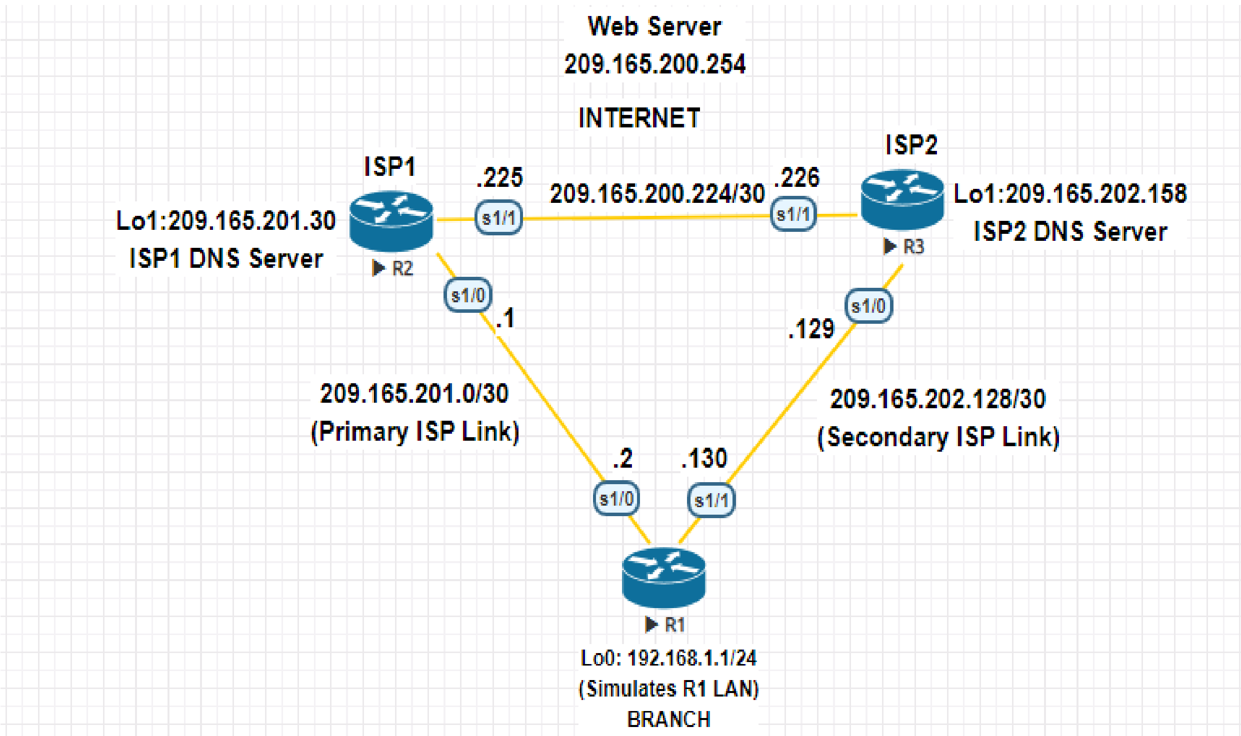
Principal/Stamp

INDEX

Practical - 1

Configure IP SLA Tracking and Path Control Topology

NETWORK TOPOLOGY



TASKS

- Configure and verify the IP SLA feature.
- Test the IP SLA tracking feature.
- Verify the configuration and operation using show and debug commands

R1

Router>enable

Router#conf t

Router(config)#hostname R1

R1(config)#interface Loopback 0

R1(config-if)#ip address 192.168.1.1 255.255.255.0

R1(config-if)#exit

R1(config)#interface s1/0

R1(config-if)#ip address 209.165.201.2 255.255.255.252

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#interface s1/1

R1(config-if)#ip address 209.165.202.130 255.255.255.252

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.201.1

R1(config)#ip sla 12

R1(config-ip-sla)#icmp-echo 209.165.201.30

R1(config-ip-sla-echo)#frequency 11

R1(config-ip-sla-echo)#exit

R1(config)#ip sla schedule 12 life forever start-time now

```
R1#sh ip sla configuration 12
IP SLAs Infrastructure Engine-III
Entry number: 12
Owner:
Tag:
Operation timeout (milliseconds): 5000
Type of operation to perform: icmp-echo
Target address/Source address: 209.165.201.30/0.0.0.0
Type Of Service parameter: 0x0
Request size (ARR data portion): 28
Verify data: No
Vrf Name:
Schedule:
   Operation frequency (seconds): 11  (not considered if randomly scheduled)
   Next Scheduled Start Time: Start Time already passed
   Group Scheduled : FALSE
   Randomly Scheduled : FALSE
   Life (seconds): Forever
   Entry Ageout (seconds): never
   Recurring (Starting Everyday): FALSE
   Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Distribution Statistics:
   Number of statistic hours kept: 2
```

```
   Number of statistic distribution buckets kept: 1
   Statistic distribution interval (milliseconds): 20
Enhanced History:
History Statistics:
   Number of history Lives kept: 0
   Number of history Buckets kept: 15
   History Filter Type: None

R1#sh ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 12
        Latest RTT: 11 milliseconds
Latest operation start time: 18:21:25 EET Thu Apr 9 2020
Latest operation return code: OK
Number of successes: 22
Number of failures: 0
Operation time to live: Forever

R1(config)#ip sla 24
R1(config-ip-sla)#icmp-echo  209.165.202.158
R1(config-ip-sla-echo)#frequency 10
R1(config-ip-sla-echo)#exit
R1(config)#ip sla schedule 24 life forever start-time now
```

```
R1#sh ip sla configuration 24
IP SLAs Infrastructure Engine-III
Entry number: 24
Owner:
Tag:
Operation timeout (milliseconds): 5000
Type of operation to perform: icmp-echo
Target address/Source address: 209.165.202.158/0.0.0.0
Type Of Service parameter: 0x0
Request size (ARR data portion): 28
Verify data: No
Vrf Name:
Schedule:
   Operation frequency (seconds): 10  (not considered if randomly scheduled)
   Next Scheduled Start Time: Start Time already passed
   Group Scheduled : FALSE
   Randomly Scheduled : FALSE
   Life (seconds): Forever
   Entry Ageout (seconds): never
   Recurring (Starting Everyday): FALSE
   Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Distribution Statistics:
   Number of statistic hours kept: 2
```

Number of statistic distribution buckets kept: 1

Statistic distribution interval (milliseconds): 20

Enhanced History:

History Statistics:

Number of history Lives kept: 0

Number of history Buckets kept: 15

History Filter Type: None

R1#sh ip sla statistics 24

IPSLAs Latest Operation Statistics

IPSLA operation id: 24

Latest RTT: 20 milliseconds

Latest operation start time: 18:33:25 EET Thu Apr 9 2020

Latest operation return code: OK

Number of successes: 16

Number of failures: 0

Operation time to live: Forever

R1(config)#no ip route 0.0.0.0 0.0.0.0 209.165.201.1

R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.201.1 5

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S*    0.0.0.0/0 [5/0] via 209.165.201.1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, Loopback0
L        192.168.1.1/32 is directly connected, Loopback0
      209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks
C        209.165.201.0/30 is directly connected, Serial1/0
L        209.165.201.2/32 is directly connected, Serial1/0
      209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks
C        209.165.202.128/30 is directly connected, Serial1/1
L        209.165.202.130/32 is directly connected, Serial1/1
```

```
R1(config)#track 1 ip sla 12 reachability
R1(config-track)#delay down 10 up 1
R1(config-track)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.201.1 2 track 1

R1(config)#track 2 ip sla 12 reachability
R1(config-track)#delay down 10 up 1
R1(config-track)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.201.1 3 track 2

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S*    0.0.0.0/0 [3/0] via 209.165.201.1
```

```
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, Loopback0
L       192.168.1.1/32 is directly connected, Loopback0
     209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.201.0/30 is directly connected, Serial1/0
L       209.165.201.2/32 is directly connected, Serial1/0
     209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.202.128/30 is directly connected, Serial1/1
L       209.165.202.130/32 is directly connected, Serial1/1

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S*    0.0.0.0/0 [5/0] via 209.165.201.1
```

```
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, Loopback0
L       192.168.1.1/32 is directly connected, Loopback0
     209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.201.0/30 is directly connected, Serial1/0
L       209.165.201.2/32 is directly connected, Serial1/0
     209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.202.128/30 is directly connected, Serial1/1
L       209.165.202.130/32 is directly connected, Serial1/1

R1#sh ip sla statistics
IPSLAs Latest Operation Statistics
IPSLA operation id: 12
        Latest RTT: NoConnection/Busy/Timeout
Latest operation start time: 19:02:29 EET Thu Apr 9 2020
Latest operation return code: Timeout
Number of successes: 227
Number of failures: 19
Operation time to live: Forever
IPSLA operation id: 24
        Latest RTT: 20 milliseconds
Latest operation start time: 19:02:35 EET Thu Apr 9 2020
Latest operation return code: OK
Number of successes: 190
```

```
Number of failures: 1
Operation time to live: Forever

R1#trace 209.165.200.254 source 192.168.1.1
Type escape sequence to abort.
Tracing the route to 209.165.200.254
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.201.1 10 msec 14 msec *

R1#sh ip sla statistics
IPSLAs Latest Operation Statistics
IPSLA operation id: 12
        Latest RTT: 10 milliseconds
Latest operation start time: 19:07:04 EET Thu Apr 9 2020
Latest operation return code: OK
Number of successes: 236
Number of failures: 35
Operation time to live: Forever
IPSLA operation id: 24
        Latest RTT: 21 milliseconds
Latest operation start time: 19:07:05 EET Thu Apr 9 2020
Latest operation return code: OK
Number of successes: 217
Number of failures: 1
```

Operation time to live: Forever

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S*    0.0.0.0/0 [3/0] via 209.165.201.1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, Loopback0
L        192.168.1.1/32 is directly connected, Loopback0
      209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks
C        209.165.201.0/30 is directly connected, Serial1/0
L        209.165.201.2/32 is directly connected, Serial1/0
      209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks
C        209.165.202.128/30 is directly connected, Serial1/1
```

L   209.165.202.130/32 is directly connected, Serial1/1

ISP1 (R2)

Router>enable

Router#conf t

Router(config)#hostname ISP1

ISP1(config)#interface Loopback0

ISP1(config-if)#description Simulated Internet Web Server

ISP1(config-if)#ip address 209.165.200.254 255.255.255.255

ISP1(config-if)#exit

ISP1(config)#interface Loopback1

ISP1(config-if)#ip address 209.165.201.30 255.255.255.255

ISP1(config-if)#exit

ISP1(config)#interface s1/0

ISP1(config-if)#ip address 209.165.201.1 255.255.255.252

ISP1(config-if)#no shutdown

ISP1(config-if)#exit

ISP1(config)#interface s1/1

ISP1(config-if)#ip address 209.165.200.225 255.255.255.252

ISP1(config-if)#no shutdown

ISP1(config-if)#exit

ISP1(config)#router eigrp 200

ISP1(config-router)#network 209.165.200.224

ISP1(config-router)#network 209.165.201.0

ISP1(config-router)#no auto-summary

ISP1(config-router)#exit

ISP1(config)#ip route 192.168.1.0 255.255.255.0 209.165.201.2

ISP1(config)#interface loopback 1

ISP1(config-if)#shut

ISP1(config)#interface loopback 1

ISP1(config-if)#no shutdown

ISP2 (R3)

Router>enable

Router#conf t

Router(config)#hostname ISP2

ISP2(config)#interface Loopback0

ISP2(config-if)#description Simulated Internet Web Server

ISP2(config-if)#ip address 209.165.200.254 255.255.255.255

ISP2(config-if)#exit

ISP2(config)#interface Loopback1

ISP2(config-if)#ip address 209.165.202.158 255.255.255.255

ISP2(config-if)#exit

ISP2(config)#interface s1/1

ISP2(config-if)#ip address 209.165.200.226 255.255.255.252

ISP2(config-if)#no shutdown

ISP2(config-if)#exit

ISP2(config)#interface s1/0

ISP2(config-if)#ip address 209.165.202.129 255.255.255.252

ISP2(config-if)#no shutdown

ISP2(config-if)#exit

ISP2(config)#router eigrp 200

ISP2(config-router)#network 209.165.200.224

ISP2(config-router)#network 209.165.202.128

ISP2(config-router)#no auto-summary

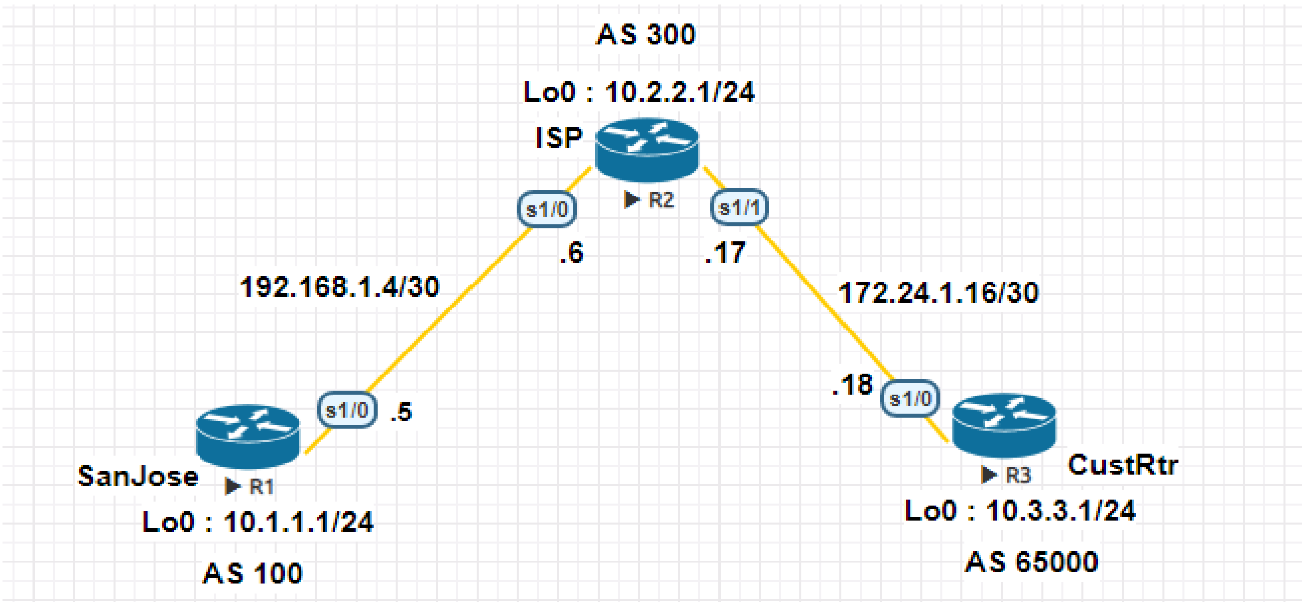
ISP2(config-router)#exit

ISP2(config)#ip route 192.168.1.0 255.255.255.0 209.165.202.130

Practical - 2

Using the AS_PATH Attribute

NETWORK TOPOLOGY



<u>TASKS</u>

- **Use BGP commands to prevent private AS numbers from being advertised to the outside world.**
- **Use the AS_PATH attribute to filter BGP routes based on their source AS numbers.**

<u>SanJose</u>

```
Router>enable
Router#conf t
Router(config)#hostname SanJose
SanJose(config)#interface Loopback0
SanJose(config-if)#ip address 10.1.1.1 255.255.255.0
SanJose(config-if)#exit
SanJose(config)#interface  Serial1/0
SanJose(config-if)#ip address 192.168.1.5 255.255.255.252
SanJose(config-if)#no shutdown
SanJose(config-if)#end
SanJose(config)#router bgp 100
SanJose(config-router)#network 10.1.1.0 mask 255.255.255.0
SanJose(config-router)#neighbor 192.168.1.6 remote-as 300
SanJose(config-router)#exit

SanJose#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
```

```
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C        10.1.1.0/24 is directly connected, Loopback0
L        10.1.1.1/32 is directly connected, Loopback0
B        10.2.2.0/24 [20/0] via 192.168.1.6, 00:05:47
B        10.3.3.0/24 [20/0] via 192.168.1.6, 00:02:13
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.4/30 is directly connected, Serial1/0
L        192.168.1.5/32 is directly connected, Serial1/0

SanJose#sh ip bgp
BGP table version is 4, local router ID is 10.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  10.1.1.0/24      0.0.0.0                  0         32768 i
```

```
 *>  10.2.2.0/24      192.168.1.6            0           0 300 i
 *>  10.3.3.0/24      192.168.1.6                       0 300 65000 i
```

```
SanJose#sh ip bgp
BGP table version is 5, local router ID is 10.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  10.1.1.0/24      0.0.0.0                  0         32768 i
 *>  10.2.2.0/24      192.168.1.6              0             0 300 i
 *>  10.3.3.0/24      192.168.1.6                            0 300 i
```

<u>ISP</u>

```
Router>enable
Router#conf t
Router(config)#hostname ISP
ISP(config)#interface Loopback0
ISP(config-if)#ip address 10.2.2.1 255.255.255.0
ISP(config-if)#exit
ISP(config)#interface Serial1/0
```

ISP(config-if)#ip address 192.168.1.6 255.255.255.252

ISP(config-if)#no shutdown

ISP(config-if)#exit

ISP(config)#interface Serial1/1

ISP(config-if)#ip address 172.24.1.17 255.255.255.252

ISP(config-if)#no shutdown

ISP(config-if)#end

ISP(config)#router bgp 300

ISP(config-router)#network 10.2.2.0 mask 255.255.255.0

ISP(config-router)#neighbor 192.168.1.5 remote-as 100

ISP(config-router)#neighbor 172.24.1.18 remote-as 65000

ISP(config)#router bgp 300

ISP(config-router)#neighbor 192.168.1.5 remove-private-as

ISP(config-router)#end

ISP#clear ip bgp * soft

ISP(config)#ip as-path access-list 1 deny ^100$

ISP(config)#ip as-path access-list 1 permit .*

ISP(config)#router bgp 300

ISP(config-router)#neighbor 172.24.1.18 filter-list 1 out

ISP(config-router)#end

```
ISP#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
B        10.1.1.0/24 [20/0] via 192.168.1.5, 00:46:41
C        10.2.2.0/24 is directly connected, Loopback0
L        10.2.2.1/32 is directly connected, Loopback0
B        10.3.3.0/24 [20/0] via 172.24.1.18, 00:43:07
      172.24.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.24.1.16/30 is directly connected, Serial1/1
L        172.24.1.17/32 is directly connected, Serial1/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.4/30 is directly connected, Serial1/0
L        192.168.1.6/32 is directly connected, Serial1/0
```

ISP#show ip bgp regexp ^100$

BGP table version is 4, local router ID is 10.2.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

```
    Network          Next Hop            Metric LocPrf Weight Path
 *>  10.1.1.0/24      192.168.1.5              0             0 100 i
```

<u>CustRtr</u>

Router>enable

Router#conf t

Router(config)#hostname CustRtr

CustRtr(config)#interface Loopback0

CustRtr(config-if)#ip address 10.3.3.1 255.255.255.0

CustRtr(config-if)#exit

CustRtr(config)#interface Serial1/0

CustRtr(config-if)#ip address 172.24.1.18 255.255.255.252

CustRtr(config-if)#no shutdown

CustRtr(config-if)#end

CustRtr(config)#router bgp 65000

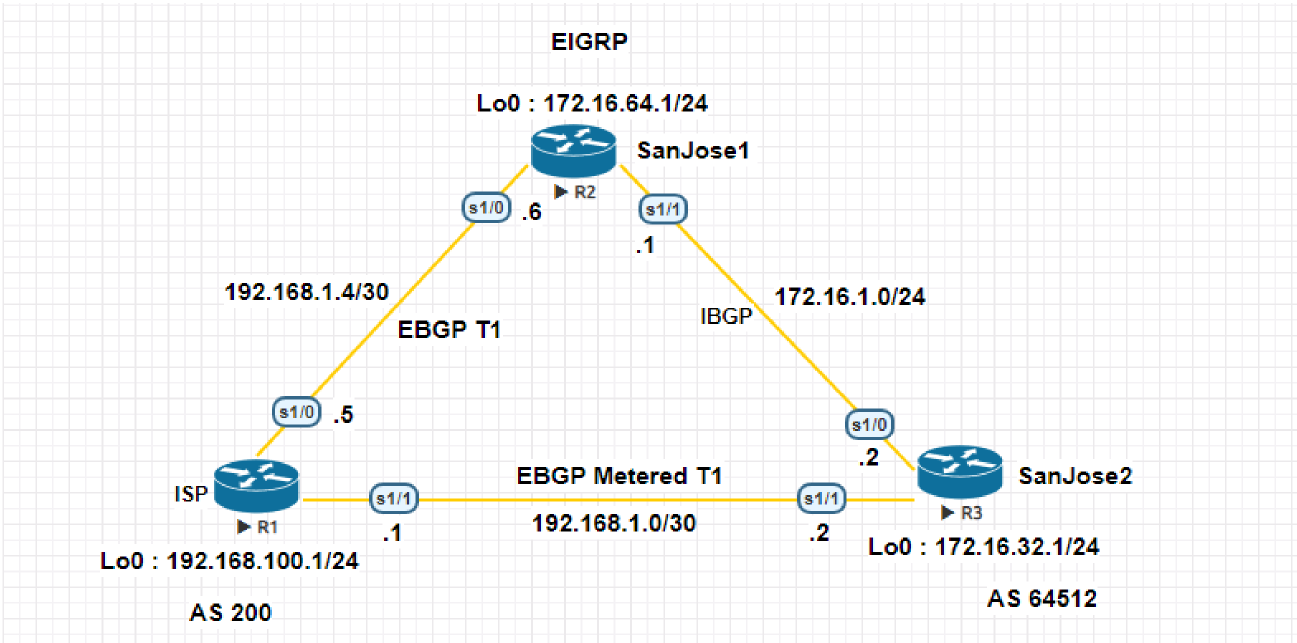
CustRtr(config-router)#network 10.3.3.0 mask 255.255.255.0

```
CustRtr(config-router)#neighbor 172.24.1.17 remote-as 300
CustRtr(config-router)#end

CustRtr#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
B        10.2.2.0/24 [20/0] via 172.24.1.17, 00:45:59
C        10.3.3.0/24 is directly connected, Loopback0
L        10.3.3.1/32 is directly connected, Loopback0
      172.24.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.24.1.16/30 is directly connected, Serial1/0
L        172.24.1.18/32 is directly connected, Serial1/0
```

Practical - 3

Configuring IBGP and EBGP Sessions, Local Preference, and MED

NETWORK TOPOLOGY



<u>TASKS</u>

- For IBGP peers to correctly exchange routing information, use the next-hop-self command with the Local-Preference and MED attributes.
- Ensure that the flat-rate, unlimited-use T1 link is used for sending and receiving data to and from the AS 200 on ISP and that the metered T1 only be used in the event that the primary T1 link has failed

R1(ISP)

Router>enable

Router#conf t

Router(config)#hostname ISP

ISP(config)#interface Loopback0

ISP(config-if)#ip address 192.168.100.1 255.255.255.0

ISP(config-if)#exit

ISP(config)#interface Serial1/0

ISP(config-if)#ip address 192.168.1.5 255.255.255.252

ISP(config-if)#no shutdown

ISP(config-if)#exit

ISP(config)#interface Serial1/1

ISP(config-if)#ip address 192.168.1.1 255.255.255.252

ISP(config-if)#no shutdown

ISP(config-if)#exit

ISP(config)#router bgp 200

ISP(config-router)#network 192.168.100.0

ISP(config-router)#neighbor 192.168.1.6 remote-as 64512

ISP(config-router)#neighbor 192.168.1.2 remote-as 64512

ISP(config-router)#exit

```
ISP#sh ip bgp
BGP table version is 3, local router ID is 192.168.100.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *   172.16.0.0       192.168.1.2              0             0 64512 i
 *>                   192.168.1.6              0             0 64512 i
 *>  192.168.100.0    0.0.0.0                  0         32768 i

ISP#ping 172.16.1.1 source 192.168.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.100.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 10/10/11 ms
ISP#ping 172.16.32.1 source 192.168.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.32.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.100.1
!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 15/15/16 ms
ISP#ping 172.16.1.2 source 192.168.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:
Packet sent with a source address of 192.168.100.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 15/17/25 ms

ISP(config)#router bgp 200
ISP(config-router)#network 192.168.1.0 mask 255.255.255.252
ISP(config-router)#network 192.168.1.4 mask 255.255.255.252
ISP(config-router)#exit
ISP#sh ip bgp
BGP table version is 5, local router ID is 192.168.100.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *   172.16.0.0       192.168.1.6              0             0 64512 i
 *>                   192.168.1.2              0             0 64512 i
 *>  192.168.1.0/30   0.0.0.0                  0         32768 i
```

```
 *>  192.168.1.4/30   0.0.0.0                0       32768 i
 *>  192.168.100.0    0.0.0.0                0       32768 i

ISP#sh ip bgp
BGP table version is 6, local router ID is 192.168.100.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  172.16.0.0       192.168.1.6             50             0 64512 i
 *                    192.168.1.2             75             0 64512 i
 *>  192.168.1.0/30   0.0.0.0                  0         32768 i
 *>  192.168.1.4/30   0.0.0.0                  0         32768 i
 *>  192.168.100.0    0.0.0.0                  0         32768 i

ISP#ping 172.16.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/10/11 ms
```

ISP#ping 172.16.1.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 20/21/25 ms

ISP#traceroute 172.16.1.1

Type escape sequence to abort.

Tracing the route to 172.16.1.1

VRF info: (vrf in name/id, vrf out name/id)

1 192.168.1.6 10 msec 10 msec *

ISP#traceroute 172.16.1.2

Type escape sequence to abort.

Tracing the route to 172.16.1.2

VRF info: (vrf in name/id, vrf out name/id)

1 192.168.1.6 10 msec 10 msec 13 msec

2 172.16.1.2 [AS 64512] 20 msec 19 msec *

R2 (SanJose1)

```
Router>enable
Router#conf t
Router(config)#hostname SanJose1
SanJose1(config)#interface Loopback0
SanJose1(config-if)#ip address 172.16.64.1 255.255.255.0
SanJose1(config-if)#ip address 172.16.64.1 255.255.255.0
SanJose1(config-if)#exit
SanJose1(config)#interface  Serial1/0
SanJose1(config-if)#ip address 192.168.1.6 255.255.255.252
SanJose1(config-if)#no shutdown
SanJose1(config-if)#exit
SanJose1(config)#interface Serial1/1
SanJose1(config-if)#ip address 172.16.1.1 255.255.255.0
SanJose1(config-if)#no shutdown
SanJose1(config-if)#exit
SanJose1(config)#router eigrp 64512
SanJose1(config-router)#network 172.16.0.0
SanJose1(config-router)#no auto-summary
SanJose1(config-router)#exit
SanJose1(config)#router bgp 64512
SanJose1(config-router)#neighbor 172.16.32.1 remote-as 64512
SanJose1(config-router)#neighbor 172.16.32.1 update-source loopback0
SanJose1(config-router)#exit
```

```
SanJose1(config)#ip route 172.16.0.0 255.255.0.0 null 0
SanJose1(config)#router bgp 64512
SanJose1(config-router)#network  172.16.0.0
SanJose1(config-router)#neighbor 192.168.1.5 remote-as 200
SanJose1(config-router)#exit
SanJose1(config)#router bgp 64512
SanJose1(config-router)#neighbor 172.16.32.1 next-hop-self
SanJose1(config-router)#exit

SanJose1#sh ip bgp
BGP table version is 5, local router ID is 172.16.64.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
            r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
            x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

    Network          Next Hop            Metric LocPrf Weight Path
 * i 172.16.0.0       172.16.32.1              0    100      0 i
 *>                   0.0.0.0                  0         32768 i
 * i 192.168.1.0/30   172.16.32.1              0    100      0 200 i
 *>                   192.168.1.5              0             0 200 i
 r i 192.168.1.4/30   172.16.32.1              0    100      0 200 i
 r>                   192.168.1.5              0             0 200 i
```

```
 * i 192.168.100.0    172.16.32.1              0    100      0 200 i
 *>                   192.168.1.5              0             0 200 i
```

SanJose1(config)#route-map PRIMARY_T1_IN permit 10

SanJose1(config-route-map)#set local-preference 160

SanJose1(config-route-map)#exit

SanJose1(config)#router bgp 64512

SanJose1(config-router)#neighbor 192.168.1.5 route-map PRIMARY_T1_IN in

SanJose1(config-router)#exit

SanJose1#clear ip bgp * soft

```
SanJose1#sh ip bgp
BGP table version is 8, local router ID is 172.16.64.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

    Network          Next Hop            Metric LocPrf Weight Path
 * i 172.16.0.0       172.16.32.1              0    100      0 i
 *>                   0.0.0.0                  0         32768 i
 *>  192.168.1.0/30   192.168.1.5              0    160      0 200 i
 r>  192.168.1.4/30   192.168.1.5              0    160      0 200 i
```

```
 *>  192.168.100.0    192.168.1.5              0    160      0 200 i

SanJose1(config)#route-map PRIMARY_T1_MED_OUT permit 10
SanJose1(config-route-map)#set Metric 50
SanJose1(config-route-map)#exit
SanJose1(config)#router bgp 64512
SanJose1(config-router)#neighbor 192.168.1.5 route-map
PRIMARY_T1_MED_OUT out
SanJose1(config-router)#exit
SanJose1(config)#exit
SanJose1#clear ip bgp * soft

SanJose1#sh ip bgp
BGP table version is 8, local router ID is 172.16.64.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 * i 172.16.0.0       172.16.32.1              0    100      0 i
 *>                   0.0.0.0                  0         32768 i
 *>  192.168.1.0/30   192.168.1.5              0    160      0 200 i
 r>  192.168.1.4/30   192.168.1.5              0    160      0 200 i
```

```
 *>  192.168.100.0    192.168.1.5             0    160      0 200 i

SanJose1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 6 subnets, 3 masks
S        172.16.0.0/16 is directly connected, Null0
C        172.16.1.0/24 is directly connected, Serial1/1
L        172.16.1.1/32 is directly connected, Serial1/1
D        172.16.32.0/24 [90/2297856] via 172.16.1.2, 01:28:25, Serial1/1
C        172.16.64.0/24 is directly connected, Loopback0
L        172.16.64.1/32 is directly connected, Loopback0
      192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
B        192.168.1.0/30 [20/0] via 192.168.1.5, 00:45:28
```

C 192.168.1.4/30 is directly connected, Serial1/0

L 192.168.1.6/32 is directly connected, Serial1/0

B 192.168.100.0/24 [20/0] via 192.168.1.5, 00:45:28

After issuing ip default-network

SanJose1(config)#ip default-network 192.168.100.0

SanJose1(config)#end

SanJose1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is 192.168.1.5 to network 192.168.100.0

S* 0.0.0.0/0 [20/0] via 192.168.1.5

172.16.0.0/16 is variably subnetted, 6 subnets, 3 masks

S 172.16.0.0/16 is directly connected, Null0

```
C       172.16.1.0/24 is directly connected, Serial1/1
L       172.16.1.1/32 is directly connected, Serial1/1
D       172.16.32.0/24 [90/2297856] via 172.16.1.2, 01:33:38, Serial1/1
C       172.16.64.0/24 is directly connected, Loopback0
L       172.16.64.1/32 is directly connected, Loopback0
     192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
B       192.168.1.0/30 [20/0] via 192.168.1.5, 00:50:41
C       192.168.1.4/30 is directly connected, Serial1/0
L       192.168.1.6/32 is directly connected, Serial1/0
B*    192.168.100.0/24 [20/0] via 192.168.1.5, 00:50:41

SanJose1#ping 192.168.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 14/15/16 ms
SanJose1#traceroute 192.168.1.2
Type escape sequence to abort.
Tracing the route to 192.168.1.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.1.5 [AS 200] 10 msec 10 msec 10 msec
  2 192.168.1.2 [AS 200] 15 msec 15 msec *
```

SanJose1#ping 192.168.1.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 9/9/11 ms

SanJose1#traceroute 192.168.1.1

Type escape sequence to abort.

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

1 192.168.1.5 [AS 200] 10 msec 11 msec *

R3 (SanJose2)

Router>en

Router#conf t

Router(config)#hostname SanJose2

SanJose2(config)#interface Loopback0

SanJose2(config-if)#ip address 172.16.32.1 255.255.255.0

SanJose2(config-if)#exit

SanJose2(config)#interface Serial1/1

SanJose2(config-if)#ip address 192.168.1.2 255.255.255.252

SanJose2(config-if)#no shutdown

SanJose2(config-if)#exit

SanJose2(config)#interface Serial1/0

SanJose2(config-if)#ip address 172.16.1.2 255.255.255.0

```
SanJose2(config-if)#no shutdown
SanJose2(config-if)#exit
SanJose2(config)#router eigrp 64512
SanJose2(config-router)#network 172.16.0.0
SanJose2(config-router)#no auto-summary
SanJose2(config-router)#exit
SanJose2(config)#router bgp 64512
SanJose2(config-router)#neighbor 172.16.64.1 remote-as 64512
SanJose2(config-router)#neighbor 172.16.64.1 update-source loopback0
SanJose2(config-router)#exit
SanJose2(config)#ip route 172.16.0.0 255.255.0.0 null 0
SanJose2(config)#router bgp 64512
SanJose2(config-router)#network  172.16.0.0
SanJose2(config-router)#neighbor 192.168.1.1 remote-as 200
SanJose2(config-router)#exit

SanJose2#sh ip bgp summary
BGP router identifier 172.16.32.1, local AS number 64512
BGP table version is 4, main routing table version 4
2 network entries using 280 bytes of memory
4 path entries using 320 bytes of memory
4/2 BGP path/bestpath attribute entries using 576 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
```

```
0 BGP filter-list cache entries using 0 bytes of memory

BGP using 1200 total bytes of memory

BGP activity 2/0 prefixes, 4/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down
State/PfxRcd

172.16.64.1     4       64512      31      32        4    0    0 00:24:41        2

192.168.1.1     4         200       8       6        4    0    0 00:01:22        1


SanJose2#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

       E1 - OSPF external type 1, E2 - OSPF external type 2

       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

       ia - IS-IS inter area, * - candidate default, U - per-user static route

       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

       a - application route

       + - replicated route, % - next hop override


Gateway of last resort is not set


      172.16.0.0/16 is variably subnetted, 6 subnets, 3 masks

S        172.16.0.0/16 is directly connected, Null0

C        172.16.1.0/24 is directly connected, Serial1/0

L        172.16.1.2/32 is directly connected, Serial1/0
```

```
C       172.16.32.0/24 is directly connected, Loopback0
L       172.16.32.1/32 is directly connected, Loopback0
D        172.16.64.0/24 [90/2297856] via 172.16.1.1, 00:08:46, Serial1/0
     192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C        192.168.1.0/30 is directly connected, Serial1/1
L       192.168.1.2/32 is directly connected, Serial1/1
B        192.168.1.4/30 [20/0] via 192.168.1.1, 00:02:19
B     192.168.100.0/24 [20/0] via 192.168.1.1, 00:07:40

SanJose2#sh ip bgp
BGP table version is 5, local router ID is 172.16.32.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 * i 172.16.0.0       172.16.64.1              0    100      0 i
 *>                   0.0.0.0                  0         32768 i
 r i 192.168.1.0/30   192.168.1.5              0    100      0 200 i
 r>                   192.168.1.1              0             0 200 i
 * i 192.168.1.4/30   192.168.1.5              0    100      0 200 i
 *>                   192.168.1.1              0             0 200 i
```

```
 * i 192.168.100.0    192.168.1.5             0    100      0 200 i
 *>                192.168.1.1             0            0 200 i

SanJose2(config)#router bgp 64512
SanJose2(config-router)#neighbor 172.16.64.1 next-hop-self
SanJose2(config-router)#exit

SanJose2#sh ip bgp
BGP table version is 5, local router ID is 172.16.32.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

    Network          Next Hop            Metric LocPrf Weight Path
 * i 172.16.0.0       172.16.64.1             0    100      0 i
 *>                0.0.0.0                 0         32768 i
 r i 192.168.1.0/30   172.16.64.1             0    100      0 200 i
 r>                192.168.1.1             0            0 200 i
 * i 192.168.1.4/30   172.16.64.1             0    100      0 200 i
 *>                192.168.1.1             0            0 200 i
 * i 192.168.100.0    172.16.64.1             0    100      0 200 i
 *>                192.168.1.1             0            0 200 i
```

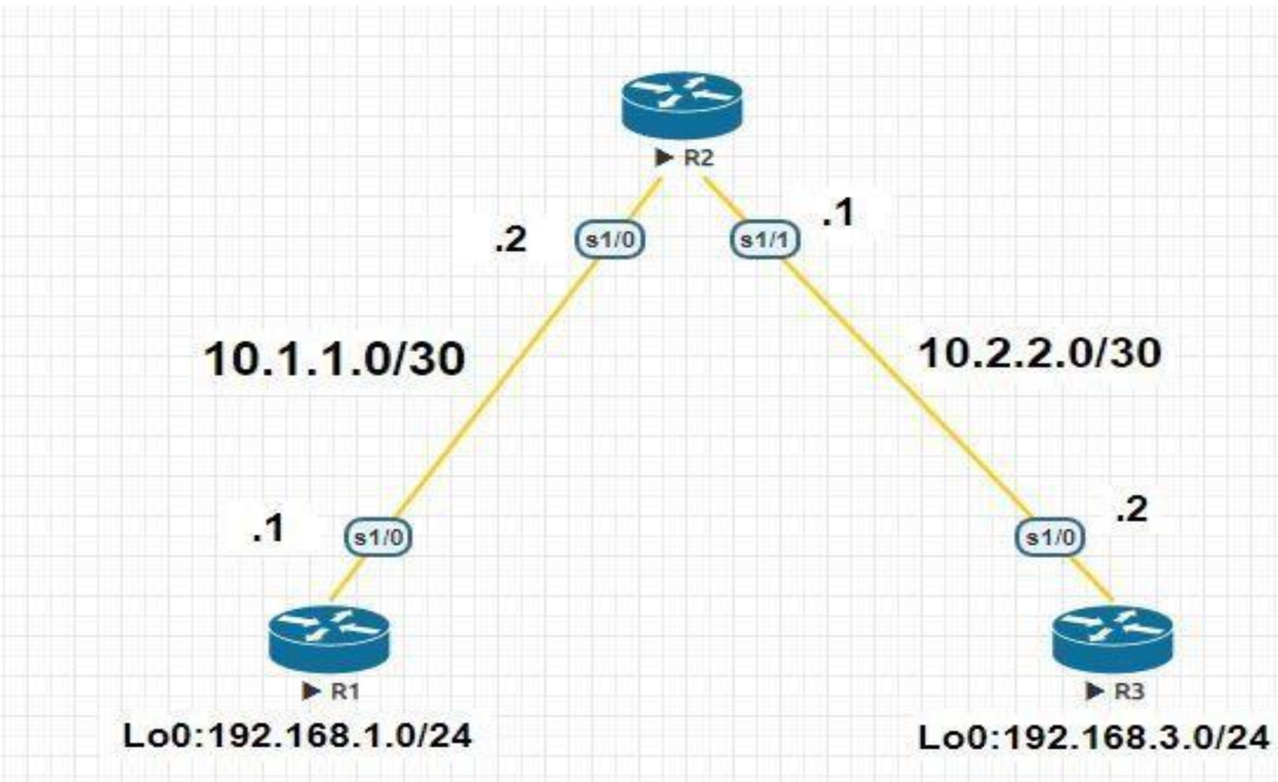
```
SanJose2(config)#route-map SECONDARY_T1_IN permit 10
SanJose2(config-route-map)#set local-preference 125
SanJose2(config-route-map)#exit
SanJose2(config)#router bgp 64512
SanJose2(config-router)#neighbor 192.168.1.1 route-map SECONDARY_T1_IN in
SanJose2(config-router)#exit
SanJose2#clear ip bgp * soft
SanJose2#sh ip bgp
BGP table version is 8, local router ID is 172.16.32.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
            r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
            x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop            Metric LocPrf Weight Path
 * i 172.16.0.0       172.16.64.1              0    100      0 i
 *>                   0.0.0.0                  0         32768 i
 r>i 192.168.1.0/30   172.16.64.1              0    160      0 200 i
 r                    192.168.1.1              0    125      0 200 i
 *>i 192.168.1.4/30   172.16.64.1              0    160      0 200 i
 *                    192.168.1.1              0    125      0 200 i
 *>i 192.168.100.0    172.16.64.1              0    160      0 200 i
 *                    192.168.1.1              0    125      0 200 i
```

```
SanJose2(config)#route-map SECONDARY_T1_MED_OUT permit 10
SanJose2(config-route-map)#set Metric 75
SanJose2(config-route-map)#exit
SanJose2(config)#router bgp 64512
SanJose2(config-router)#$2.168.1.1 route-map SECONDARY_T1_MED_OUT out
SanJose2(config-router)#end
SanJose2#clear ip bgp * soft
SanJose2#sh ip bgp
BGP table version is 8, local router ID is 172.16.32.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
        r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
        x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found
  Network          Next Hop            Metric LocPrf Weight Path
 * i 172.16.0.0       172.16.64.1           0    100      0 i
 *>                   0.0.0.0               0         32768 i
 r>i 192.168.1.0/30   172.16.64.1           0    160      0 200 i
 r                    192.168.1.1           0    125      0 200 i
 *>i 192.168.1.4/30   172.16.64.1           0    160      0 200 i
 *                    192.168.1.1           0    125      0 200 i
 *>i 192.168.100.0    172.16.64.1           0    160      0 200 i
 *                    192.168.1.1           0    125      0 200 i
```

Practical - 4

Secure the Management Plane

NETWORK TOPOLOGY



TASKS

- **Secure Management Access**
- **Configure enhanced username password security**
- **Enable AAA RADIUS authentication**
- **Enable Secure Remote Management**

R1

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname R1

R1(config)#interface Loopback 0

***Dec 19 07:53:42.473: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up**

R1(config-if)#ip address 192.168.1.1 255.255.255.0

R1(config-if)#exit

R1(config)#interface s1/0

R1(config-if)#ip address 10.1.1.1 255.255.255.252

R1(config-if)#no shutdown

***Dec 19 07:57:21.998: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up**

***Dec 19 07:57:22.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up**

R1(config-if)#exit

R1(config)#exit

Configure static routes

a. **On R1, configure a default static route to ISP.**

R1(config)# ip route 0.0.0.0 0.0.0.0 10.1.1.2

R1#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B – BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default,U - per-user static route o - ODR, P - periodic downloaded static route,H - NHRP,l - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is 10.1.1.2 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 10.1.1.2

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.1.1.0/30 is directly connected, Serial1/0

L 10.1.1.1/32 is directly connected, Serial1/0

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.0/24 is directly connected, Loopback0

L 192.168.1.1/32 is directly connected, Loopback0

Secure management access

R1(config)#security passwords min-length 10

R1(config)#enable secret class12345

R1(config)#line console 0

R1(config-line)#password ciscoconpass

R1(config-line)#exec-timeout 5 0

R1(config-line)#login

R1(config-line)#logging synchronous

R1(config-line)#exit

R1(config)#line vty 0 4

R1(config-line)#password ciscovtypass

R1(config-line)#exec-timeout 5 0

R1(config-line)#login

R1(config-line)#exit

R1(config)#line aux 0

R1(config-line)#no exec

R1(config-line)#end

R1(config)#service password-encryption

R1(config)#banner motd $Unauthorized access strictly prohibited!$

R1(config)#exit

Configure enhanced username password security

R1(config)#username JR-ADMIN secret class12345

R1(config)#username ADMIN secret class54321

R1(config)#line console 0

R1(config-line)#login local

R1(config-line)#end

R1(config)#line vty 0 4

R1(config-line)#login local

R1(config-line)#end

Enabling AAA RADIUS Authentication with Local User for Backup

```
R1(config)# aaa new-model
R1(config)# radius server RADIUS-1
R1(config-radius-server)# address ipv4 192.168.1.101
R1(config-radius-server)# key RADIUS-1-pa55w0rd
R1(config-radius-server)# exit
R1(config)# radius server RADIUS-2
R1(config-radius-server)# address ipv4 192.168.1.102
R1(config-radius-server)# key RADIUS-2-pa55w0rd
R1(config-radius-server)# exit
R1(config)# aaa group server radius RADIUS-GROUP
R1(config-sg-radius)# server name RADIUS-1
R1(config-sg-radius)# server name RADIUS-2
R1(config-sg-radius)# exit
R1(config)# aaa authentication login default group RADIUS-GROUP local
R1(config)# aaa authentication login TELNET-LOGIN group RADIUS-GROUP local-case
R1(config)# line vty 0 4
R1(config-line)# login authentication TELNET-LOGIN
```

R1(config-line)# exit

R2

Router>enable

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname R2

R2(config)#interface s1/0

R2(config-if)#ip address 10.1.1.2 255.255.255.252

R2(config-if)#no shutdown

***Dec 19 08:01:10.279: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up**

***Dec 19 08:01:11.279: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up**

R2(config-if)#exit

R2(config)#interface s1/1

R2(config-if)#ip address 10.2.2.1 255.255.255.252

R2(config-if)#no shutdown

***Dec 19 08:02:33.002: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up**

***Dec 19 08:02:34.009: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up**

R2(config-if)#exit

R2(config)#exit

Configure static routes

a. **On R2, configure two static routes.**

R2(config)# ip route 192.168.1.0 255.255.255.0 10.1.1.1

R2(config)# ip route 192.168.3.0 255.255.255.0 10.2.2.2

R2#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B – BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2 ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks

C 10.1.1.0/30 is directly connected, Serial1/0

L 10.1.1.2/32 is directly connected, Serial1/0

C 10.2.2.0/30 is directly connected, Serial1/1

L 10.2.2.1/32 is directly connected, Serial1/1

S 192.168.1.0/24 [1/0] via 10.1.1.1

S 192.168.3.0/24 [1/0] via 10.2.2.2

Secure management access

R2(config)#security passwords min-length 10

R2(config)#enable secret class12345

R2(config)#line console 0

R2(config-line)#password ciscoconpass

R2(config-line)#exec-timeout 5 0

R2(config-line)#login

R2(config-line)#logging synchronous

R2(config-line)#exit

R2(config)#line vty 0 4

R2(config-line)#password ciscovtypass

R2(config-line)#exec-timeout 5 0

R2(config-line)#login

R2(config-line)#exit

R2(config)#line aux 0

R2(config-line)#no exec

R2(config-line)#end

R2(config)#service password-encryption

R2(config)#banner motd $Unauthorized access strictly prohibited!$

R2(config)#exit

Configure enhanced username password security

R2(config)#username JR-ADMIN secret class12345

R2(config)#username ADMIN secret class54321

R2(config)#line console 0

R2(config-line)#login local

R2(config-line)#end

R2(config)#line vty 0 4

R2(config-line)#login local

R2(config-line)#end

Enabling AAA RADIUS Authentication with Local User for Backup

R2(config)# aaa new-model
R2(config)# radius server RADIUS-1
R2(config-radius-server)# address ipv4 192.168.1.101
R2(config-radius-server)# key RADIUS-1-pa55w0rd
R2(config-radius-server)# exit
R2(config)# radius server RADIUS-2
R2(config-radius-server)# address ipv4 192.168.1.102
R2(config-radius-server)# key RADIUS-2-pa55w0rd
R2(config-radius-server)# exit
R2(config)# aaa group server radius RADIUS-GROUP
R2(config-sg-radius)# server name RADIUS-1
R2(config-sg-radius)# server name RADIUS-2
R2(config-sg-radius)# exit
R2(config)# aaa authentication login default group RADIUS-GROUP local
R2(config)# aaa authentication login TELNET-LOGIN group RADIUS-GROUP local-case
R2(config)# line vty 0 4
R2(config-line)# login authentication TELNET-LOGIN

R2(config-line)# exit

R3

Router>enable

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname R3

R3(config)#interface loopback 0

***Dec 19 08:07:50.079: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up**

R3(config-if)#ip address 192.168.3.1 255.255.255.0

R3(config-if)#exit

R3(config)#interface s1/0

R3(config-if)#ip address 10.2.2.2 255.255.255.252

R3(config-if)#no shutdown

R3(config-if)#exit

***Dec 19 08:09:26.986: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up**

***Dec 19 08:09:27.996: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up**

R3(config)#end

Configure static routes

a. **On R3, configure a default static route to ISP.**

   R3(config)# ip route 0.0.0.0 0.0.0.0 10.2.2.1

   R3#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B – BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is 10.2.2.1 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 10.2.2.1

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.2.2.0/30 is directly connected, Serial1/0

L 10.2.2.2/32 is directly connected, Serial1/0

192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.3.0/24 is directly connected, Loopback0

L 192.168.3.1/32 is directly connected, Loopback0

<u>Secure management access</u>

R3(config)#security passwords min-length 10

R3(config)#enable secret class12345

R3(config)#line console 0

R3(config-line)#password ciscoconpass

R3(config-line)#exec-timeout 5 0

R3(config-line)#login

R3(config-line)#logging synchronous

R3(config-line)#exit

R3(config)#line vty 0 4

R3(config-line)#password ciscovtypass

R3(config-line)#exec-timeout 5 0

R3(config-line)#login

R3(config-line)#exit

R3(config)#line aux 0

R3(config-line)#no exec

R3(config-line)#end

R3(config)#service password-encryption

R3(config)#banner motd $Unauthorized access strictly prohibited!$

<u>Configure enhanced username password security</u>

R3(config)#username JR-ADMIN secret class12345

R3(config)#username ADMIN secret class54321

R3(config)#line console 0

R3(config-line)#login local

R3(config-line)#exit

R3(config)#line vty 0 4

R3(config-line)#login local

R3(config-line)#exit

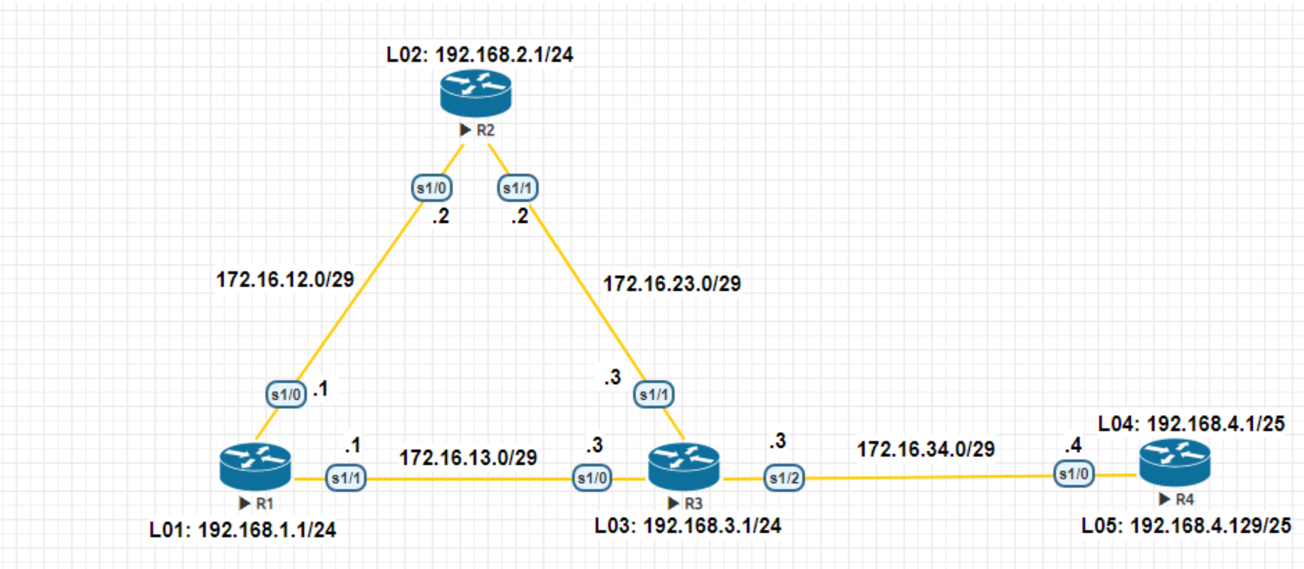
<u>Enabling AAA RADIUS Authentication with Local User for Backup</u>

```
R3(config)# aaa new-model
R3(config)# radius server RADIUS-1
R3(config-radius-server)# address ipv4 192.168.1.101
R3(config-radius-server)# key RADIUS-1-pa55w0rd
R3(config-radius-server)# exit
R3(config)# radius server RADIUS-2
R3(config-radius-server)# address ipv4 192.168.1.102
R3(config-radius-server)# key RADIUS-2-pa55w0rd
R3(config-radius-server)# exit
R3(config)# aaa group server radius RADIUS-GROUP
R3(config-sg-radius)# server name RADIUS-1
R3(config-sg-radius)# server name RADIUS-2
R3(config-sg-radius)# exit
R3(config)# aaa authentication login default group RADIUS-GROUP local
R3(config)# aaa authentication login TELNET-LOGIN group RADIUS-GROUP local-case
R3(config)# line vty 0 4
R3(config-line)# login authentication TELNET-LOGIN
R3(config-line)# exit
```

Practical - 5

Configure and Verify Path Control Using PBR

NETWORK TOPOLOGY



<u>TASKS</u>

- **Configure and verify policy-based routing.**
- **Select the required tools and commands to configure policy-based routing operations.**
- **Verify the configuration and operation by using the proper show and debug commands**

R1

Router>enable

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname R1

R1(config)#interface Lo1

R1(config-if)#ip address 192.168.1.1 255.255.255.0

R1(config-if)#exit

R1(config)#interface s1/0

R1(config-if)#ip address 172.16.12.1 255.255.255.248

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#interface s1/1

R1(config-if)#ip address 172.16.13.1 255.255.255.248

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#router eigrp 100

R1(config-router)#network 192.168.1.0

R1(config-router)#network 172.16.12.0

R1(config-router)#network 172.16.13.0

R1(config-router)#no auto-summary

R1(config-router)#exit

```
R1#sh ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(100)
H   Address          Interface       Hold Uptime   SRTT   RTO  Q  Seq
                                     (sec)         (ms)       Cnt Num
1   172.16.13.3      Se1/1             14 00:04:43   11   100  0  10
0   172.16.12.2      Se1/0             12 00:07:05   19   114  0  8
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C        172.16.12.0/29 is directly connected, Serial1/0
L        172.16.12.1/32 is directly connected, Serial1/0
C        172.16.13.0/29 is directly connected, Serial1/1
L        172.16.13.1/32 is directly connected, Serial1/1
```

```
D       172.16.23.0/29 [90/2681856] via 172.16.13.3, 00:08:31, Serial1/1
                       [90/2681856] via 172.16.12.2, 00:08:31, Serial1/0
D       172.16.34.0/29 [90/2681856] via 172.16.13.3, 00:08:31, Serial1/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, Loopback1
L       192.168.1.1/32 is directly connected, Loopback1
D    192.168.2.0/24 [90/2297856] via 172.16.12.2, 00:08:31, Serial1/0
D    192.168.3.0/24 [90/2297856] via 172.16.13.3, 00:08:31, Serial1/1
     192.168.4.0/25 is subnetted, 2 subnets
D       192.168.4.0 [90/2809856] via 172.16.13.3, 00:05:15, Serial1/1
D       192.168.4.128 [90/2809856] via 172.16.13.3, 00:05:15, Serial1/1
```

R2

```
Router>enable
Router#conf t
Router(config)#hostname R2
R2(config)#interface Lo2
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#exit
R2(config)#interface s1/0
R2(config-if)#ip address 172.16.12.2 255.255.255.248
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface s1/1
```

R2(config-if)#ip address 172.16.23.2 255.255.255.248

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#router eigrp 100

R2(config-router)#network 192.168.2.0

R2(config-router)#network 172.16.12.0

R2(config-router)#network 172.16.23.0

R2(config-router)#no auto-summary

R2#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
1	172.16.23.3	Se1/1	12	00:05:23	12	100	0	11
0	172.16.12.1	Se1/0	12	00:07:45	22	132	0	8

<u>R3</u>

Router>enable

Router#conf t

Router(config)#hostname R3

R3(config)#interface Lo3

R3(config-if)#ip address 192.168.3.1 255.255.255.0

R3(config-if)#exit

R3(config)#interface s1/0

R3(config-if)#ip address 172.16.13.3 255.255.255.248

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#interface s1/1

R3(config-if)#ip address 172.16.23.3 255.255.255.248

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#interface s1/2

R3(config-if)#ip address 172.16.34.3 255.255.255.248

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#router eigrp 100

R3(config-router)#network 192.168.3.0

R3(config-router)#network 172.16.13.0

R3(config-router)#network 172.16.23.0

R3(config-router)#network 172.16.34.0

R3(config-router)#no auto-summary

```
R3#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H   Address          Interface         Hold Uptime   SRTT   RTO  Q  Seq
                                       (sec)         (ms)       Cnt Num
2   172.16.34.4      Se1/2               14 00:03:09   15   100  0  3
1   172.16.13.1      Se1/0               14 00:06:25   21   126  0  9
0   172.16.23.2      Se1/1               13 00:06:25   20   120  0  9


R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 7 subnets, 2 masks
D        172.16.12.0/29 [90/2681856] via 172.16.23.2, 00:16:48, Serial1/1
                        [90/2681856] via 172.16.13.1, 00:16:48, Serial1/0
```

```
C      172.16.13.0/29 is directly connected, Serial1/0
L      172.16.13.3/32 is directly connected, Serial1/0
C      172.16.23.0/29 is directly connected, Serial1/1
L      172.16.23.3/32 is directly connected, Serial1/1
C      172.16.34.0/29 is directly connected, Serial1/2
L      172.16.34.3/32 is directly connected, Serial1/2
D   192.168.1.0/24 [90/2297856] via 172.16.13.1, 00:16:48, Serial1/0
D    192.168.2.0/24 [90/2297856] via 172.16.23.2, 00:16:48, Serial1/1
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.3.0/24 is directly connected, Loopback3
L       192.168.3.1/32 is directly connected, Loopback3
    192.168.4.0/25 is subnetted, 2 subnets
D       192.168.4.0 [90/2297856] via 172.16.34.4, 00:13:32, Serial1/2
D       192.168.4.128 [90/2297856] via 172.16.34.4, 00:13:32, Serial1/2

R3(config)#ip access-list standard PBR-ACL
R3(config-std-nacl)#remark ACL matches R4 LAN B traffic
R3(config-std-nacl)#permit 192.168.4.128 0.0.0.127
R3(config-std-nacl)#exit
R3(config)#route-map R3-to-R1 permit
R3(config-route-map)#match ip address PBR-ACL
R3(config-route-map)#set ip next-hop 172.16.13.1
R3(config-route-map)#end
```

R3(config)#int s1/2

R3(config-if)#ip policy route-map R3-to-R1

R3(config-if)#exit

R3#sh route-map

route-map R3-to-R1, permit, sequence 10

Match clauses:

ip address (access-lists): PBR-ACL

Set clauses:

ip next-hop 172.16.13.1

Policy routing matches: 0 packets, 0 bytes

R3(config)#access-list 1 permit 192.168.4.0 0.0.0.255

R4

Router>enable

Router#conf t

Router(config)#hostname R4

R4(config)#interface lo4

R4(config-if)#ip address 192.168.4.1 255.255.255.128

R4(config-if)#exit

R4(config)#interface lo5

R4(config-if)#ip address 192.168.4.129 255.255.255.128

R4(config-if)#exit

R4(config)#interface s1/0

R4(config-if)#ip address 172.16.34.4 255.255.255.248

```
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#router eigrp 100
R4(config-router)#network 192.168.4.0
R4(config-router)#network 172.16.34.0
R4(config-router)#no auto-summary

R4#sh ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(100)
H   Address          Interface        Hold Uptime   SRTT   RTO  Q  Seq
                                      (sec)         (ms)       Cnt Num
0   172.16.34.3      Se1/0              14 00:04:07   25   150  0  9
```

Before Route Maps

```
R4#traceroute 192.168.1.1 source 192.168.4.1
Type escape sequence to abort.
Tracing the route to 192.168.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.34.3 13 msec 11 msec 10 msec
  2 172.16.13.1 20 msec 17 msec *
```

R4#traceroute 192.168.1.1 source 192.168.4.129

Type escape sequence to abort.

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

1 172.16.34.3 15 msec 10 msec 10 msec

2 172.16.13.1 19 msec 24 msec *

<u>After Route Maps</u>

R4#traceroute 192.168.1.1 source 192.168.4.1

Type escape sequence to abort.

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

1 172.16.34.3 11 msec 10 msec 10 msec

2 172.16.13.1 21 msec 22 msec *

R4#traceroute 192.168.1.1 source 192.168.4.129

Type escape sequence to abort.

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

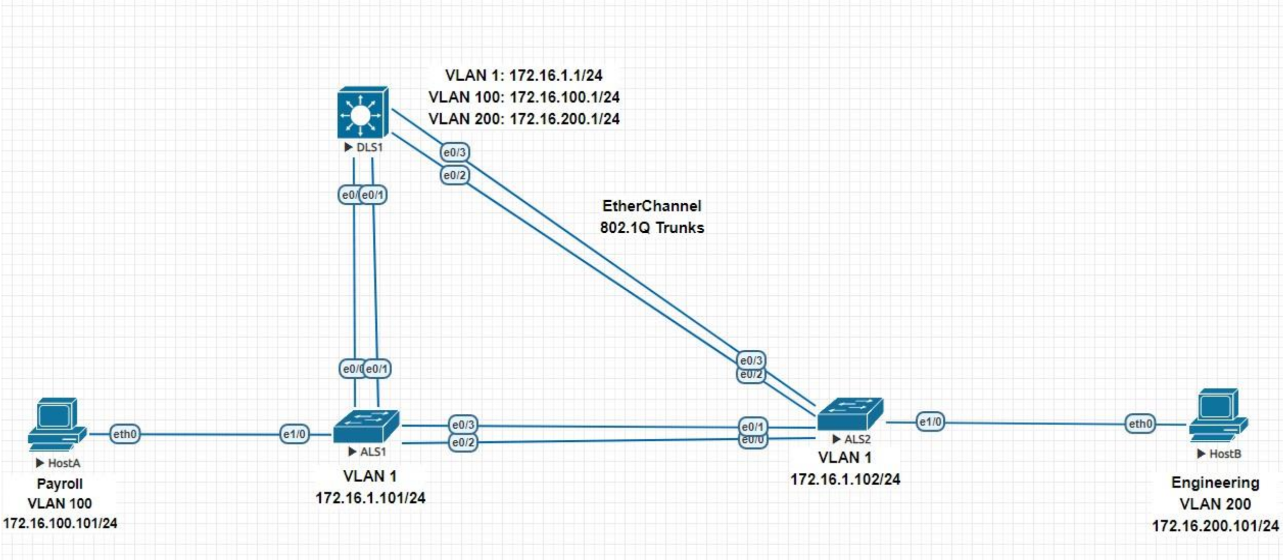
1 172.16.34.3 10 msec 10 msec 10 msec

2 172.16.13.1 18 msec 18 msec *

Practical - 6

IP Service Level Agreements and Remote SPAN in a Campus Environment

NETWORK TOPOLOGY



TASKS

- **Configure trunking, VTP, and SVIs**
- **Implement IP SLAs to monitor various network performance characteristics**
- **Implement Remote Span**

DLS1

Switch>en

Switch#conf t

Switch(config)#hostname DLS1

DLS1(config)#interface vlan 1

DLS1(config-if)#ip address 172.16.1.1 255.255.255.0

DLS1(config-if)#no shutdown

DLS1(config-if)#exit

Configure the trunks and EtherChannel from DLS1 to ALS1.

DLS1(config)#interface range e0/0-1

DLS1(config-if-range)#switchport trunk encapsulation dot1q

DLS1(config-if-range)#switchport mode trunk

DLS1(config-if-range)#channel-group 1 mode desirable

Creating a port-channel interface Port-channel 1

DLS1(config-if-range)#exit

Configure the trunks and EtherChannel from DLS1 to ALS2.

DLS1(config)#interface range e0/2-3

DLS1(config-if-range)#switchport trunk encapsulation dot1q

DLS1(config-if-range)#switchport mode trunk

DLS1(config-if-range)#channel-group 2 mode desirable

Creating a port-channel interface Port-channel 2

DLS1(config-if-range)#exit

Configure VTP on DLS1 and create VLANs 100 and 200 for the domain

DLS1(config)#vtp domain SWPOD

Changing VTP domain name from NULL to SWPOD

DLS1(config)#vtp version 2

DLS1(config)#vlan 100

DLS1(config-vlan)#name Payroll

DLS1(config-vlan)#exit

DLS1(config)#vlan 200

DLS1(config-vlan)#name Engineering

DLS1(config-vlan)#exit

On DLS1, create the SVIs for VLANs 100 and 200. Note that the corresponding Layer 2 VLANs must be configured for the Layer 3 SVIs to activate

DLS1(config)#interface vlan 100

DLS1(config-if)#ip address 172.16.100.1 255.255.255.0

DLS1(config-if)#no shutdown

DLS1(config-if)#exit

DLS1(config)#interface vlan 200

DLS1(config-if)#ip address 172.16.200.1 255.255.255.0

DLS1(config-if)#no shutdown

DLS1(config-if)#exit

The ip routing command is also needed to allow the DLS1 switch to act as a Layer 3 device to route between these VLANs. Because the VLANs are all considered directly connected, a routing protocol is not needed at this time. The default configuration on 3560 switches is no ip routing.

DLS1(config)#ip routing

```
DLS1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C        172.16.1.0/24 is directly connected, Vlan1
L        172.16.1.1/32 is directly connected, Vlan1
C        172.16.100.0/24 is directly connected, Vlan100
L        172.16.100.1/32 is directly connected, Vlan100
C        172.16.200.0/24 is directly connected, Vlan200
L        172.16.200.1/32 is directly connected, Vlan200
```

Configure the Cisco IOS IP SLA source to measure network performance

DLS1(config)#ip sla 1

DLS1(config-ip-sla)#icmp-echo 172.16.100.101

DLS1(config-ip-sla-echo)#exit

DLS1(config)#ip sla 2

DLS1(config-ip-sla)#icmp-echo 172.16.200.101

DLS1(config-ip-sla-echo)#exit

DLS1(config)#ip sla 3

DLS1(config-ip-sla)#udp-jitter 172.16.1.101 5000

DLS1(config-ip-sla-jitter)#exit

DLS1(config)#ip sla 4

DLS1(config-ip-sla)#udp-jitter 172.16.1.102 5000

DLS1(config-ip-sla-jitter)#exit

DLS1(config)#ip sla schedule 1 life forever start-time now

DLS1(config)#ip sla schedule 2 life forever start-time now

DLS1(config)#ip sla schedule 3 life forever start-time now

DLS1(config)#ip sla schedule 4 life forever start-time now

Monitor IP SLAs operations

DLS1#show ip sla configuration 1

IP SLAs Infrastructure Engine-III

Entry number: 1

Owner:

Tag:

Operation timeout (milliseconds): 5000

Type of operation to perform: icmp-echo

Target address/Source address: 172.16.100.101/0.0.0.0

Type Of Service parameter: 0x0

Request size (ARR data portion): 28

Data pattern: 0xABCDABCD

Verify data: No

Vrf Name:

Schedule:

Operation frequency (seconds): 60 (not considered if randomly scheduled)

Next Scheduled Start Time: Start Time already passed

Group Scheduled : FALSE

Randomly Scheduled : FALSE

Life (seconds): Forever

Entry Ageout (seconds): never

Recurring (Starting Everyday): FALSE

Status of entry (SNMP RowStatus): Active

Threshold (milliseconds): 5000

Distribution Statistics:

Number of statistic hours kept: 2

Number of statistic distribution buckets kept: 1

Statistic distribution interval (milliseconds): 20

Enhanced History:

History Statistics:

Number of history Lives kept: 0

Number of history Buckets kept: 15

History Filter Type: None

DLS1#show ip sla configuration 3

IP SLAs Infrastructure Engine-III

Entry number: 3

Owner:

Tag:

Operation timeout (milliseconds): 5000

Type of operation to perform: udp-jitter

Target address/Source address: 172.16.1.101/0.0.0.0

Target port/Source port: 5000/0

Type Of Service parameter: 0x0

Request size (ARR data portion): 32

Packet Interval (milliseconds)/Number of packets: 20/10

Verify data: No

Vrf Name:

Control Packets: enabled

Schedule:

Operation frequency (seconds): 60 (not considered if randomly scheduled)

Next Scheduled Start Time: Start Time already passed

Group Scheduled : FALSE

Randomly Scheduled : FALSE

```
  Life (seconds): Forever
  Entry Ageout (seconds): never
  Recurring (Starting Everyday): FALSE
  Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Distribution Statistics:
  Number of statistic hours kept: 2
  Number of statistic distribution buckets kept: 1
  Statistic distribution interval (milliseconds): 20
Enhanced History:
Percentile:

DLS1#show ip sla application
     IP Service Level Agreements
Version: Round Trip Time MIB 2.2.0, Infrastructure Engine-III

Supported Operation Types:
     icmpEcho, path-echo, path-jitter, udpEcho, tcpConnect, http
     dns, udpJitter, dhcp, ftp, lsp Group, lspPing, lspTrace
     pseudowirePing, udpApp, wspApp, mcast, generic

Supported Features:
     IPSLAs Event Publisher
```

IP SLAs low memory water mark: 225778552

Estimated system max number of entries: 165365

Estimated number of configurable operations: 165241

Number of Entries configured : 4

Number of active Entries : 4

Number of pending Entries : 0

Number of inactive Entries : 0

Time of last change in whole IP SLAs: *14:08:46.139 EET Sat Apr 11 2020

DLS1#show ip sla statistics 1

IPSLAs Latest Operation Statistics

IPSLA operation id: 1

Latest RTT: 1 milliseconds

Latest operation start time: 14:34:23 EET Sat Apr 11 2020

Latest operation return code: OK

Number of successes: 26

Number of failures: 1

Operation time to live: Forever

DLS1#show ip sla statistics 3

IPSLAs Latest Operation Statistics

IPSLA operation id: 3

Type of operation: udp-jitter

Latest RTT: 1 milliseconds

Latest operation start time: 14:34:36 EET Sat Apr 11 2020

Latest operation return code: OK

RTT Values:

Number Of RTT: 10 RTT Min/Avg/Max: 1/1/2 milliseconds

Latency one-way time:

Number of Latency one-way Samples: 6

Source to Destination Latency one way Min/Avg/Max: 0/0/1 milliseconds

Destination to Source Latency one way Min/Avg/Max: 0/0/1 milliseconds

Jitter Time:

Number of SD Jitter Samples: 9

Number of DS Jitter Samples: 9

Source to Destination Jitter Min/Avg/Max: 0/1/1 milliseconds

Destination to Source Jitter Min/Avg/Max: 0/1/1 milliseconds

Over Threshold:

Number Of RTT Over Threshold: 0 (0%)

Packet Loss Values:

Loss Source to Destination: 0

Source to Destination Loss Periods Number: 0

Source to Destination Loss Period Length Min/Max: 0/0

Source to Destination Inter Loss Period Length Min/Max: 0/0

Loss Destination to Source: 0

Destination to Source Loss Periods Number: 0

Destination to Source Loss Period Length Min/Max: 0/0

Destination to Source Inter Loss Period Length Min/Max: 0/0

Out Of Sequence: 0 Tail Drop: 0

Packet Late Arrival: 0 Packet Skipped: 0

Voice Score Values:

Calculated Planning Impairment Factor (ICPIF): 0

Mean Opinion Score (MOS): 0

Number of successes: 27

Number of failures: 0

Operation time to live: Forever

<u>Configure Remote Span</u>

DLS1(config)#vlan 100

DLS1(config-vlan)#remote-span

DLS1(config-vlan)#exit

DLS1(config)#monitor session 1 source interface e0/0 both

DLS1(config)# monitor session 1 destination remote vlan 100

<u>ALS1</u>

Switch>en

Switch#conf t

Switch(config)#hostname ALS1

ALS1(config)#interface vlan 1

ALS1(config-if)#ip address 172.16.1.101 255.255.255.0

ALS1(config-if)#no shutdown

ALS1(config-if)#exit

ALS1(config)#ip default-gateway 172.16.1.1

Configure the trunks and EtherChannel between ALS1 and DLS1

ALS1(config)#interface range e0/0-1

ALS1(config-if-range)# switchport trunk encapsulation dot1q

ALS1(config-if-range)#switchport mode trunk

ALS1(config-if-range)#channel-group 1 mode desirable

Creating a port-channel interface Port-channel 1

ALS1(config-if-range)#exit

Configure the trunks and EtherChannel between ALS1 and ALS2

ALS1(config)#interface range e0/2-3

ALS1(config-if-range)#switchport trunk encapsulation dot1q

ALS1(config-if-range)#switchport mode trunk

ALS1(config-if-range)#channel-group 2 mode desirable

Creating a port-channel interface Port-channel 2

Configure VTP on ALS1

ALS1(config)#vtp mode client

Setting device to VTP Client mode for VLANS.

ALS1(config)#int e1/0

ALS1(config-if)#switchport mode access

ALS1(config-if)#switchport access vlan 100

ALS1(config-if)#exit

Configure Cisco IOS IP SLA responders.

ALS1(config)#ip sla responder

ALS1(config)#ip sla responder udp-echo ipaddress 172.16.1.1 port 5000

ALS1#show ip sla responder

General IP SLA Responder on Control port 1967

General IP SLA Responder on Control V2 port 1167

General IP SLA Responder is: Enabled

Number of control message received: 16 Number of errors: 0

Recent sources:

172.16.1.1 [14:23:36.259 EET Sat Apr 11 2020]

172.16.1.1 [14:22:36.257 EET Sat Apr 11 2020]

172.16.1.1 [14:21:36.255 EET Sat Apr 11 2020]

172.16.1.1 [14:20:36.256 EET Sat Apr 11 2020]

172.16.1.1 [14:19:36.258 EET Sat Apr 11 2020]

Recent error sources:

Number of control v2 message received: 0 Number of errors: 0

Recent sources:

Recent error sources:

Permanent Port IP SLA Responder

Permanent Port IP SLA Responder is: Enabled

udpEcho Responder:

IP Address	**Port**
172.16.1.1	**5000**

ALS2

Switch>en

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#hostname ALS2

ALS2(config)#interface vlan 1

ALS2(config-if)#ip address 172.16.1.102 255.255.255.0

ALS2(config-if)#no shutdown

ALS2(config-if)#exit

ALS2(config)#ip default-gateway 172.16.1.1

Configure the trunks and EtherChannel between ALS2 and ALS1

ALS2(config)#interface range e0/0-1

ALS2(config-if-range)#switchport trunk encapsulation dot1q

ALS2(config-if-range)#switchport mode trunk

ALS2(config-if-range)#channel-group 2 mode desirable

Creating a port-channel interface Port-channel 2

ALS2(config-if-range)#exit

Configure the trunks and EtherChannel between ALS2 and DLS1

ALS2(config)#interface range e0/2-3

ALS2(config-if-range)#switchport trunk encapsulation dot1q

ALS2(config-if-range)#switchport mode trunk

ALS2(config-if-range)#channel-group 1 mode desirable

Creating a port-channel interface Port-channel 1

ALS2(config-if-range)#exit

Configure VTP on ALS2

ALS2(config)#vtp mode client

Setting device to VTP Client mode for VLANS

ALS2(config)#int e1/0

ALS2(config-if)#switchport mode access

ALS2(config-if)#switchport access vlan 200

ALS2(config-if)#exit

Configure Cisco IOS IP SLA responders.

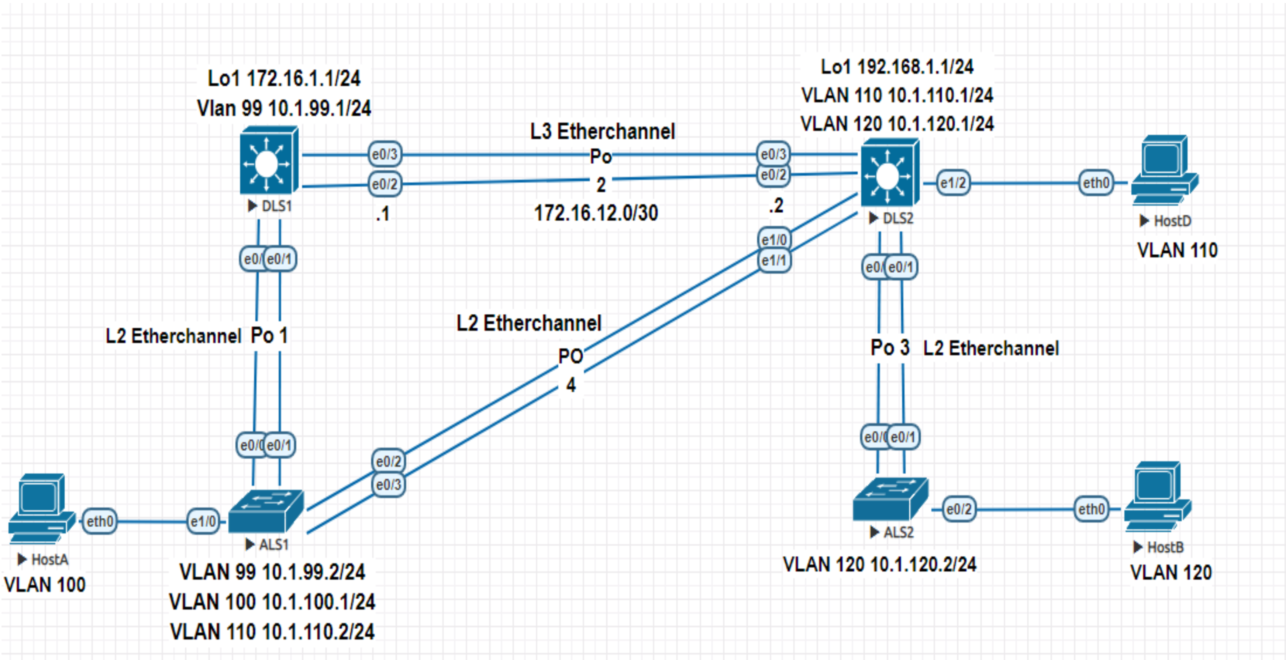
ALS2(config)#ip sla responder

ALS2(config)#ip sla responder udp-echo ipaddress 172.16.1.1 port 5000

Practical - 7

Inter-VLAN Routing

NETWORK TOPOLOGY



TASKS

- **Implement a Layer 3 EtherChannel**
- **Implement Static Routing**
- **Implement Inter-Vlan Routing**

DLS1

Switch>enable

Switch#conf t

Switch(config)#hostname DLS1

DLS1(config)#interface loopback 1

DLS1(config-if)#ip address 172.16.1.1 255.255.255.0

DLS1(config-if)#exit

DLS1(config)#interface vlan 99

DLS1(config-if)#ip address 10.1.99.1 255.255.255.0

DLS1(config-if)#no shutdown

Implement a Layer 3 EtherChannel

DLS1(config)#int range e0/2-3

DLS1(config-if-range)#no switchport

DLS1(config-if-range)#no ip address

DLS1(config-if-range)#channel-group 2 mode on

Creating a port-channel interface Port-channel 2

DLS1(config-if-range)#exit

DLS1(config)#interface port-channel 2

DLS1(config-if)#ip address 172.16.12.1 255.255.255.252

DLS1(config-if)#end

DLS1(config)#int range e0/0-1

DLS1(config-if-range)#switchport trunk encapsulation dot1q

DLS1(config-if-range)#switchport mode trunk

```
DLS1(config-if-range)#channel-group 1 mode desirable
Creating a port-channel interface Port-channel 1
DLS1(config-if-range)#end

DLS1#sh interfaces trunk
Port      Mode         Encapsulation  Status        Native vlan
Po1       on           802.1q         trunking      1

Port      Vlans allowed on trunk
Po1       1-4094

Port      Vlans allowed and active in management domain
Po1       1,99

Port      Vlans in spanning tree forwarding state and not pruned
Po1       1,99
```

Implement Static Routing

DLS1(config)#ip routing

DLS1(config)#ip route 192.168.1.0 255.255.255.252 172.16.12.2

DLS1(config)# ip route 192.168.1.0 255.255.255.0 10.1.120.1

DLS1(config)# ip route 192.168.1.0 255.255.255.0 10.1.110.1

DLS1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.1.99.0/24 is directly connected, Vlan99

L 10.1.99.1/32 is directly connected, Vlan99

172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks

C 172.16.1.0/24 is directly connected, Loopback1

```
L       172.16.1.1/32 is directly connected, Loopback1
C       172.16.12.0/30 is directly connected, Port-channel2
L       172.16.12.1/32 is directly connected, Port-channel2
    192.168.1.0/30 is subnetted, 1 subnets
S       192.168.1.0 [1/0] via 172.16.12.2
```

DLS2

```
Switch>en
Switch#conf t
Switch(config)#hostname DLS2
DLS2(config)#interface loopback 1
DLS2(config-if)#ip address 192.168.1.1 255.255.255.0
DLS2(config-if)#exit
DLS2(config)#interface vlan 110
DLS2(config-if)#ip address 10.1.110.1 255.255.255.0
DLS2(config-if)#no shutdown
DLS2(config-if)#exit
DLS2(config)#interface vlan 120
DLS2(config-if)#ip address 10.1.120.1 255.255.255.0
DLS2(config-if)#no shutdown
DLS2(config-if)#exit
```

Implement a Layer 3 EtherChannel

```
DLS2(config)#interface range e0/2-3
DLS2(config-if-range)#no switchport
DLS2(config-if-range)#no ip
DLS2(config-if-range)#no ip address
DLS2(config-if-range)#channel-group 2 mode on
Creating a port-channel interface Port-channel 2
DLS2(config-if-range)#exit
DLS2(config)#interface port-channel 2
DLS2(config-if)#ip address 172.16.12.2 255.255.255.252
DLS2(config-if)#end
DLS2(config)#interface range e0/0-1
DLS2(config-if-range)#switchport trunk encapsulation dot1q
DLS2(config-if-range)#switchport mode trunk
DLS2(config-if-range)#channel-group 3 mode desirable
Creating a port-channel interface Port-channel 3
DLS2(config-if-range)#exit
DLS2(config)#interface range e1/0-1
DLS2(config-if-range)#switchport trunk encapsulation dot1q
DLS2(config-if-range)#switchport mode trunk
DLS2(config-if-range)#channel-group 4 mode desirable
Creating a port-channel interface Port-channel 4
DLS2(config-if-range)#end
```

DLS2#sh interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Po3	on	802.1q	trunking	1
Po4	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Po3	1-4094
Po4	1-4094

Port	Vlans allowed and active in management domain
Po3	1,110,120
Po4	1,110,120

Port	Vlans in spanning tree forwarding state and not pruned
Po3	1,110,120
Po4	1,110,120

Implement Static Routing

DLS2(config)#ip routing

DLS2(config)#ip route 172.16.1.0 255.255.255.252 172.16.12.1

DLS2(config)# ip route 172.16.1.0 255.255.255.0 10.1.99.1

Configure the host ports for the appropriate VLANs according to the diagram

DLS2(config)#interface e1/2

DLS2(config-if)#switchport mode access

DLS2(config-if)#switchport access vlan 110

DLS2#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks

C 10.1.110.0/24 is directly connected, Vlan110

L 10.1.110.1/32 is directly connected, Vlan110

C 10.1.120.0/24 is directly connected, Vlan120

L 10.1.120.1/32 is directly connected, Vlan120

172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks

S 172.16.1.0/30 [1/0] via 172.16.12.1

C 172.16.12.0/30 is directly connected, Port-channel2

L 172.16.12.2/32 is directly connected, Port-channel2

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.0/24 is directly connected, Loopback1

L 192.168.1.1/32 is directly connected, Loopback1

ALS1

Switch>en

Switch#conf t

Switch(config)#hostname ALS1

ALS1(config)#ip default-gateway 10.1.99.1

ALS1(config)#ip default-gateway 10.1.110.1

ALS1(config)#ip default-gateway 10.1.100.2

Implement a Layer 3 EtherChannel

ALS1(config)#int range e0/0-1

ALS1(config-if-range)#switchport trunk encapsulation dot1q

ALS1(config-if-range)#switchport mode trunk

ALS1(config-if-range)#channel-group 1 mode desirable

Creating a port-channel interface Port-channel 1

ALS1(config-if-range)#exit

ALS1(config)#int range e0/2-3

ALS1(config-if-range)#switchport trunk encapsulation dot1q

```
ALS1(config-if-range)#switchport mode trunk
ALS1(config-if-range)#channel-group 4 mode desirable
Creating a port-channel interface Port-channel 4
ALS1(config-if-range)#end

ALS1#sh etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator
        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
        A - formed by Auto LAG
Number of channel-groups in use: 2
Number of aggregators:           2
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         PAgP      Et0/0(P)    Et0/1(P)
4      Po4(SU)         PAgP      Et0/2(P)    Et0/3(P)
```

Configure the host ports for the appropriate VLANs according to the diagram

ALS1(config)#interface e1/0

ALS1(config-if)#switchport mode access

ALS1(config-if)#switchport access vlan 100

ALS2

Switch>en

Switch#conf t

Switch(config)#hostname ALS2

ALS2(config)#ip default-gateway 10.1.120.1

Implement a Layer 3 EtherChannel

ALS2(config)#int range e0/0-1

ALS2(config-if-range)#switchport trunk encapsulation dot1q

ALS2(config-if-range)#switchport mode trunk

ALS2(config-if-range)#channel-group 3 mode desirable

Creating a port-channel interface Port-channel 3

ALS2(config-if-range)#end

```
ALS2#sh etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
3      Po3(SU)         PAgP      Et0/0(P)    Et0/1(P)
```

Configure the host ports for the appropriate VLANs according to the diagram

ALS2(config)#interface e0/2

ALS2(config-if)#switchport mode access

ALS2(config-if)#switchport access vlan 120

HOST A

VPCS> ip 10.1.100.1 255.255.255.0 10.1.100.2

HOST B

VPCS> ip 10.1.120.2 255.255.255.0 10.1.120.1

HOST D

VPCS> ip 10.1.110.2 255.255.255.0 10.1.110.1